Southeast Louisiana Pipeline Infrastructure

Port Nickel Comprehensive Analysis

Braithwaite, Plaquemines Parish, Louisiana

Site: 387 Acres, Mississippi River Mile 80 (East Bank)

Project: MASABB Site Assessment

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Executive Summary

Purpose

This comprehensive report analyzes oil and gas pipeline infrastructure in Southeast Louisiana to evaluate connectivity opportunities for the Port Nickel site (387 acres, Braithwaite, Plaquemines Parish). The analysis identifies existing and planned pipeline systems within the region, assesses proximity to the site, and evaluates strategic positioning within Louisiana's expanding energy infrastructure corridor.

Key Findings

Regional Context

- Southeast Louisiana hosts **12+ major natural gas transmission pipelines** driven by LNG export growth (15.3 Bcf/d operational, expanding to 28+ Bcf/d by 2028)
- St. James Parish (60 miles upriver) serves as Louisiana's primary **crude oil terminal hub**
- All recent pipeline construction (2023-2027) is LNG-driven; **8 major natural gas projects** under development
- This demand surge requires **$5–10 billion in new pipeline infrastructure investment** through 2030

Site Proximity

- **Gulf South Pipeline** (major intrastate gas system): 10-25 miles estimated
- **Harvest CAM System:** <5 miles - serves Belle Chasse crude oil terminal (Harvest Midstream)
- **Transco, Colonial, Collins:** 15-30 miles (East Bank corridor)
- **Plaquemines LNG feed pipelines:** 50-75 miles south (same parish)

Strategic Position

- East Bank location provides direct access to major pipeline corridor **without river crossings**
- Positioned between New Orleans/Baton Rouge refining complex (north) and Plaquemines LNG terminals (south)
- Same parish as Plaquemines LNG (Venture Global) - largest LNG growth driver in region

Plaquemines LNG Feed Infrastructure

- Gator Express Pipeline (~2.0 Bcf/d header pipeline) serves Plaquemines LNG in same parish
- East Lateral Xpress (0.725 Bcf/d, Columbia Gulf/TC Energy) operational 2025, ties into Gator Express

- Plaquemines receives feedgas from three sources: TETCO, Tennessee Gas Pipeline, Columbia Gulf
- Plaquemines LNG feedgas averaged 2.68 Bcf/d in July 2025, reaching peak of 2.92 Bcf/d on July 13, 2025; averaged 3+ Bcf/d by August 2025

STRATEGIC OPPORTUNITY

Port Nickel's East Bank location positions the site along established pipeline corridors serving Louisiana's LNG export infrastructure. The site sits within a region experiencing rapid natural gas pipeline expansion, creating potential interconnection opportunities for facilities requiring pipeline access.

Primary opportunity is natural gas interconnection via Gulf South Pipeline system or other regional gas transmission infrastructure. Proximity to Harvest CAM crude terminal demonstrates existing midstream infrastructure in immediate vicinity.

1. Regional Pipeline Infrastructure

1.1 The Gulf Coast's Most Active Energy Infrastructure Corridor

Southeast Louisiana represents one of North America's most concentrated energy infrastructure corridors. The region hosts an unparalleled density of natural gas transmission pipelines, crude oil terminals, refined products systems, and LNG export facilities. This convergence stems from:

- **Offshore Gulf of Mexico Production:** Over 15% of U.S. natural gas production comes from offshore platforms, landing onshore in coastal Louisiana parishes.
- **Haynesville Shale Proximity:** Louisiana provides the shortest path to tidewater for 15+ Bcf/d of dry gas production from the Haynesville (North Louisiana / East Texas).
- **LNG Export Boom:** Louisiana hosts 6 operational LNG export terminals (15.3 Bcf/d) and 5 major projects under construction or approved (adding 13+ Bcf/d by 2028).
- **Refining & Petrochemical Complex:** The Baton Rouge to New Orleans corridor contains multiple refineries (900,000+ BPD combined capacity within 50 miles of Port Nickel) and chemical plants requiring substantial gas and NGL feedstocks.
- **Mississippi River Strategic Value:** Deep-water access and barge connectivity make River Mile 80 (Port Nickel location) a natural logistics node.

> **REGIONAL PIPELINE COUNT**
>
> **12+ major interstate and intrastate natural gas transmission systems** cross Southeast Louisiana, with aggregate capacity exceeding 50 Bcf/d. These include Transco (largest U.S. gas pipeline), Texas Eastern Transmission, Tennessee Gas Pipeline, Gulf South Pipeline, and multiple LNG-dedicated feed systems.

The Port Nickel site sits at Mississippi River Mile 80 on the East Bank in Plaquemines Parish — directly between the New Orleans/Baton Rouge industrial corridor (north) and the rapidly expanding Plaquemines LNG terminal complex (south). This geographic position places the 387-acre site within proximity to multiple pipeline systems and positions it along an established energy corridor that is currently experiencing historic levels of infrastructure investment.

1.2 Interstate & Intrastate Pipeline Systems

Southeast Louisiana is served by both **interstate pipelines** (subject to FERC regulation, offering open-access transmission) and **intrastate pipelines** (Louisiana PSC jurisdiction, more flexible commercial arrangements).

Major Interstate Natural Gas Transmission Systems

Pipeline System	Operator	Capacity	Proximity to Site	Key Characteristics
Transcontinental Gas Pipe Line (Transco)	Williams Companies	~15 Bcf/d	15–30 miles (East Bank)	Nation's largest gas pipeline; Haynesville to Gulf Coast to Northeast markets; Station 65 compressor station in St. Bernard Parish
Texas Eastern Transmission (TETCO)	Enbridge	~9 Bcf/d	25–40 miles	Texas/Louisiana to Northeast; offshore platform connections; feeds Plaquemines LNG
Tennessee Gas Pipeline (TGP)	Kinder Morgan	~9 Bcf/d	20–35 miles	Gulf Coast to Northeast; serves Plaquemines LNG via Southwest Lateral
Columbia Gulf Transmission	TC Energy	~3.1 Bcf/d	30–50 miles	Operates East Lateral Xpress to Plaquemines LNG; connects to Gator Express
Natural Gas Pipeline of America (NGPL)	Kinder Morgan	~4 Bcf/d	40–60 miles	Permian Basin to Gulf Coast; provides supply diversity

Major Intrastate Natural Gas Systems

Pipeline System	Operator	Capacity	Proximity to Site	Key Characteristics
Gulf South Pipeline	Boardwalk Pipeline Partners	~6 Bcf/d	10–25 miles (estimated)	Louisiana's largest intrastate system; extensive network across South Louisiana; connects to most major demand centers
Louisiana Energy Gateway (LEG)	Williams / Enbridge	1.44 Bcf/d	30+ miles	Haynesville to Cameron LNG (operational 2024)
Gulf Run Pipeline	Energy Transfer	1.65 Bcf/d	30+ miles	Haynesville to Golden Pass LNG (operational 2024)

GULF SOUTH PIPELINE — PRIMARY INTERCONNECTION CANDIDATE

Gulf South Pipeline is the most logical interconnection target for the Port Nickel site. As Louisiana's largest intrastate system, it provides flexible commercial arrangements (not subject to FERC open-access requirements), extensive connectivity across South Louisiana, and established presence in the Plaquemines Parish / Greater New Orleans area.

Boardwalk Pipeline Partners (Gulf South operator) has demonstrated willingness to support industrial development through pipeline laterals and interconnection projects. Preliminary discussions with Boardwalk would be a critical component of Phase 1 infrastructure assessment.

1.3 Midstream Crude & Products Infrastructure

While natural gas pipelines dominate recent infrastructure development, Southeast Louisiana also hosts extensive crude oil and refined products pipeline systems. These systems serve the region's refining complex and connect Gulf Coast production to inland markets.

Major Crude Oil Systems

- **Louisiana Offshore Oil Port (LOOP):** World's only deepwater oil port, 18 miles offshore; 1.2 million BPD throughput capacity; 62 million barrels storage at Clovelly salt dome; connected to St. James

Parish via 48-inch, 130-mile pipeline

- **Genesis Energy Offshore Systems:** Poseidon, Eugene Island, Odyssey, CHOPS provide combined 400,000+ BPD offshore crude collection
- **St. James Terminal Hub:** 60 miles upriver from Port Nickel; receives Permian Basin crude via multiple long-haul systems; primary storage and distribution point for Louisiana refineries
- **Harvest CAM System:** Serves Belle Chasse crude oil terminal (Harvest Midstream); <5 miles from Port Nickel site

Refined Products Pipelines

System	Operator	Capacity	Route
Colonial Pipeline	Colonial Pipeline Co.	2.5 million BPD	Gulf Coast to Southeast U.S.; delivers gasoline, diesel, jet fuel
Plantation Pipeline	Kinder Morgan	720,000 BPD	Louisiana to Washington DC; refined products
Collins Pipeline	Marathon/Valero	190,000 BPD	South Louisiana refined products distribution

HARVEST CAM SYSTEM PROXIMITY

The closest confirmed pipeline infrastructure to the Port Nickel site is the Harvest CAM crude gathering system, which serves the Belle Chasse crude oil terminal operated by Harvest Midstream. This terminal is located <5 miles from the Port Nickel site and demonstrates existing midstream infrastructure presence in the immediate vicinity.

2. Site-Proximate Infrastructure

2.1 Pipeline Proximity Analysis

The Port Nickel site benefits from East Bank positioning along Louisiana's primary energy corridor. While precise pipeline locations require Louisiana DNR GIS data and operator system maps, the following proximity estimates are based on regional pipeline mapping, FERC filings, and operator disclosures:

<5 mi

HARVEST CAM SYSTEM

Crude gathering to Belle Chasse terminal

10–25 mi

GULF SOUTH PIPELINE

Intrastate gas system (estimated)

15–30 mi

TRANSCO / COLONIAL / COLLINS

East Bank corridor systems

50–75 mi

PLAQUEMINES LNG PIPELINES

Gator Express, East Lateral Xpress (same parish)

Priority Systems for Phase 1 Mapping

1. **Gulf South Pipeline Network:** Request detailed system maps from Boardwalk Pipeline Partners showing all mainline segments, laterals, and compressor stations within 25-mile radius of Port Nickel.
2. **Transco Station 65 Infrastructure:** Transco's St. Bernard Parish compressor station represents potential interconnection point; obtain station location and system configuration.
3. **Harvest CAM System:** Confirm exact route and capacity of Harvest's crude gathering system serving Belle Chasse terminal.
4. **Louisiana DNR Pipeline Data:** Obtain official state pipeline GIS data for 25-mile radius around River Mile 80 (East Bank).

2.2 Direct Connection Opportunities

A pipeline lateral from Port Nickel to Gulf South Pipeline or another regional transmission system would require:

- **Right-of-Way Acquisition:** 10-25 miles of pipeline corridor (depending on connection point); likely following existing road or utility corridors where possible
- **Pipeline Construction:** 12-inch to 24-inch diameter lateral (sizing depends on volume requirements and system pressure)
- **Meter Station & Interconnection:** Custody transfer measurement, pressure regulation, and tie-in to transmission system
- **Regulatory Approvals:** Louisiana PSC approval (if intrastate connection) or FERC approval (if interstate connection)

> **REGULATORY DISTINCTION: INTERSTATE VS. INTRASTATE**
>
> **Interstate Connection (FERC jurisdiction):** Requires FERC Section 7(c) certificate; open-access requirements; more complex regulatory process but guaranteed capacity rights.
>
> **Intrastate Connection (Louisiana PSC):** Simpler state-level approval; more flexible commercial arrangements; Gulf South Pipeline (intrastate) offers this pathway.
>
> For a facility requiring pipeline access, an **intrastate connection to Gulf South Pipeline** likely provides the most straightforward regulatory and commercial pathway.

3. Planned and Development Projects

3.1 Pipeline Additions: 2023–2027

All major pipeline construction activity in Southeast Louisiana over the 2023-2027 period is driven by LNG export terminal expansion. The region is experiencing unprecedented natural gas infrastructure investment, with 8 major pipeline projects either recently completed or under active construction:

Project	Operator	Capacity	Status	LNG Terminal Served
Gator Express Pipeline	Venture Global / Gulf Companies	~2.0 Bcf/d	Operational	Plaquemines LNG Phase 1 & 2 (Plaquemines Parish)
East Lateral Xpress	Columbia Gulf / TC Energy	0.725 Bcf/d	Operational (2025)	Plaquemines LNG (ties into Gator Express)
Louisiana Energy Gateway (LEG)	Williams / Enbridge	1.44 Bcf/d	Operational (2024)	Cameron LNG Train 4 (Cameron Parish)
Gulf Run Pipeline	Energy Transfer	1.65 Bcf/d	Operational (2024)	Golden Pass LNG (Sabine Pass)
Texas-LA Energy Pathway	Williams	1.45 Bcf/d	Under construction	Plaquemines LNG brownfield expansion
Louisiana Energy Access Project (LEAP)	DT Midstream	1.15 Bcf/d	Operational (2024)	Plaquemines LNG / Venture Global CP2
NG3 Pipeline	Momentum Midstream	1.65 Bcf/d	Under construction	Venture Global CP2 (Plaquemines Parish)
Hugh Brinson Expansion	Energy Transfer	0.85 Bcf/d	Operational (2024)	Golden Pass LNG support

Combined new capacity: 10.9+ Bcf/d of natural gas transmission infrastructure added 2023-2027

> **INVESTMENT SCALE**
>
> These 8 projects represent an estimated **$5–10 billion in new pipeline infrastructure investment** across Louisiana. This level of capital deployment reflects the scale of LNG export growth and the premium placed on reliable gas supply to meet firm LNG offtake commitments.

3.2 Regional Expansion Drivers

LNG Export Terminal Development

Louisiana's LNG export capacity is expanding from 15.3 Bcf/d (operational as of 2025) to 28+ Bcf/d by 2028, driven by:

- **Plaquemines LNG Phase 1 & 2 (Venture Global):** 3.25 Bcf/d, operational 2024-2025
- **Venture Global CP2 (Plaquemines Parish):** 3.25 Bcf/d, under construction, target 2026-2027
- **Plaquemines LNG Brownfield Expansion:** 30+ MTPA additional capacity announced November 2025 (FERC CP26-27); requires new Kosciusko Junction Pipeline Project
- **Golden Pass LNG:** 2.6 Bcf/d, operational 2024-2025
- **Cameron LNG Train 4:** 0.7 Bcf/d, operational 2024

Haynesville Shale Production Growth

The Haynesville Shale (North Louisiana / East Texas) produces 15+ Bcf/d of dry natural gas and remains one of the lowest-cost U.S. gas supply basins. All new Louisiana LNG terminals are designed to receive Haynesville gas via dedicated pipeline systems. Proximity to Haynesville (150-250 miles from Port Nickel) provides supply reliability and basis differentials that favor Louisiana LNG projects over competitors in other regions.

Industrial and Power Demand Growth

Beyond LNG, Southeast Louisiana is experiencing incremental gas demand from:

- **Data center development:** Multiple hyperscale data center projects announced in Louisiana; substantial power demand growth
- **Petrochemical expansion:** Ethylene crackers, ammonia plants, methanol facilities adding 2-3 Bcf/d incremental demand
- **Refinery and industrial consumption:** Baton Rouge to New Orleans corridor consumes 2-3 Bcf/d for process heating, hydrogen production, and power generation

4. Regional Gas Flow Patterns

4.1 LNG Export Growth & Demand

LNG export terminals represent the single largest driver of natural gas demand growth in Louisiana. The state's operational LNG capacity of 15.3 Bcf/d (as of 2025) is expanding to 28+ Bcf/d by 2028 — an increase equivalent to 20% of total U.S. gas production.

15.3 Bcf/d CURRENT LOUISIANA LNG CAPACITY (2025)	**28+ Bcf/d** PROJECTED CAPACITY BY 2028
12.7 Bcf/d NEW CAPACITY UNDER CONSTRUCTION	**~85%** CURRENT UTILIZATION RATE

Plaquemines Parish — the same parish where Port Nickel is located — is home to the largest concentration of LNG growth in the United States:

- **Plaquemines LNG Phase 1 & 2:** 3.25 Bcf/d operational capacity (20 MTPA)
- **Venture Global CP2:** 3.25 Bcf/d under construction (20 MTPA); expected operational 2026-2027
- **Plaquemines LNG Brownfield Expansion:** 30+ MTPA additional capacity (FERC CP26-27 filed November 2025)

Combined: Plaquemines Parish could host 70+ MTPA (11+ Bcf/d) of LNG export capacity by 2030, making it the single largest LNG export hub in North America.

4.2 Emerging Demand Beyond LNG

While LNG dominates infrastructure investment, other demand sources are creating additional pipeline utilization and potential interconnection opportunities:

Data Center Power Demand

- Louisiana has attracted multiple hyperscale data center announcements due to low power costs and renewable energy availability
- Natural gas combined-cycle power plants provide firm, dispatchable generation to support data center loads
- Incremental gas demand estimated at 0.5-1.0 Bcf/d by 2030 from data center-driven power growth

Petrochemical and Industrial Consumption

- Baton Rouge to New Orleans refining and petrochemical corridor consumes 2-3 Bcf/d for process heating, hydrogen production, and power generation
- Three refineries within 20-50 miles of Port Nickel site: combined capacity exceeding 900,000 BPD
- Ethylene crackers, ammonia plants, and methanol facilities add incremental gas demand beyond LNG

4.3 Supply Sources & Directional Flow

Louisiana's gas supply comes from three primary sources:

1. **Haynesville Shale:** 15+ Bcf/d production from North Louisiana and East Texas; 150-250 miles from Port Nickel; connected via LEG, NG3, LEAP, Gulf Run, and other systems
2. **Offshore Gulf of Mexico:** 2-3 Bcf/d production from offshore platforms; connects via onshore Louisiana gathering systems
3. **Interstate Imports:** Gas from Appalachia (Marcellus/Utica) and Midwest flows through Louisiana to LNG facilities via Transco, Texas Eastern, and Tennessee Gas Pipeline

> **FLOW PATTERN SHIFTS**
>
> The Plaquemines LNG ramp has already shifted flows on regional pipelines. Midcontinent Express Pipeline (MEP) started delivering gas to Tennessee Gas Pipeline to meet Plaquemines demand, while volumes to Transco declined. This reallocation of capacity to support LNG demand demonstrates how new load centers reshape pipeline economics — and why strategically located storage and interconnection facilities gain value.

5. Site Implications and Strategic Assessment

5.1 Network Integration Opportunity

The Port Nickel site's location at River Mile 80 (East Bank, Plaquemines Parish) provides several strategic advantages for pipeline-dependent industrial development:

- **East Bank Positioning:** Direct access to major pipeline corridors without requiring Mississippi River crossings (which add cost, complexity, and regulatory hurdles)
- **Between Two Major Demand Centers:** Positioned between the New Orleans/Baton Rouge industrial corridor (north) and Plaquemines LNG terminals (south), creating potential for bidirectional gas flow and storage value
- **Same Parish as Largest LNG Growth:** Plaquemines Parish hosts 70+ MTPA of LNG capacity (operational and under construction), creating pipeline density and interconnection options
- **Proximity to Gulf South Pipeline:** Louisiana's largest intrastate system (10-25 miles estimated) provides most flexible interconnection pathway
- **Existing Midstream Infrastructure:** Harvest CAM crude system (<5 miles) demonstrates infrastructure presence and right-of-way precedent

POTENTIAL USE CASES

Pipeline interconnection at Port Nickel could support multiple facility types:

- **Salt Cavern Storage:** Natural gas or NGL storage serving LNG load-balancing or industrial supply reliability (subject to geological feasibility)
- **Industrial Gas Supply:** Power generation, hydrogen production, petrochemical feedstock, or other gas-consuming processes
- **Logistics & Transloading:** Barge-to-truck or barge-to-rail transfer facilities for liquid hydrocarbons
- **Pipeline Junction Station:** Interconnection point between systems, potentially with compression or measurement services

5.2 Investment Requirements

Developing pipeline connectivity from Port Nickel to regional transmission infrastructure would require capital investment across multiple categories:

Investment Category	Estimated Range	Key Variables
Pipeline Lateral Construction	$30M – $100M	Distance (10-25 miles), diameter (12-24 inches), terrain, right-of-way costs
Interconnection & Metering	$5M – $15M	Custody transfer equipment, pressure regulation, tie-in costs
Regulatory & Permitting	$2M – $5M	FERC vs. Louisiana PSC process, environmental reviews, legal counsel
Engineering & Design	$3M – $8M	Route surveys, geotechnical studies, design work
Site Infrastructure	$10M – $50M	Compression (if required), storage tanks, control systems, utilities

Total Estimated Pipeline Connectivity Investment: $50M – $178M

If the project involves salt cavern storage development, add:

- **Salt Cavern Development:** $150M – $400M (depending on number and size of caverns, depth, leaching timeline)
- **Surface Facilities for Storage:** $20M – $80M (compressors, dehydration, injection/withdrawal equipment)

Total Project Cost with Storage: $220M – $658M

5.3 Risk Factors

Geological Uncertainty

Salt dome feasibility at Braithwaite location unconfirmed. Requires seismic surveys and geological assessment before storage development can proceed.

Capital Intensity

$200–600M total project cost (salt cavern development + surface infrastructure + pipeline lateral). Requires patient capital and experienced midstream partner.

Regulatory Complexity

Interstate pipeline connections require FERC approval (2-3 year timeline). Salt cavern storage requires Louisiana DNR permits. Wetlands and coastal zone management add layers of regulatory review.

Policy Risk

Federal LNG export policy uncertainty affects post-2028 timelines. Commodity price volatility impacts storage economics. CP2 DOE review uncertainty.

6. Phased Development Pathway

Given the capital intensity, regulatory complexity, and geological uncertainty, a phased approach is recommended to de-risk the project and establish feasibility before committing to full development.

Phase 1: Infrastructure Mapping & Preliminary Assessment

Objective

Obtain definitive pipeline location data, confirm proximity estimates, and identify specific interconnection candidates.

Actions

- **Louisiana DNR Pipeline GIS Data:** Request official state pipeline mapping for 25-mile radius around River Mile 80 (East Bank)
- **Gulf South Pipeline System Maps:** Contact Boardwalk Pipeline Partners to request detailed system maps showing mainline segments, laterals, and compressor stations
- **Operator Outreach:** Preliminary discussions with potential pipeline partners:
    - Boardwalk (Gulf South Pipeline)
    - Williams (Transco)
    - Venture Global (Plaquemines LNG infrastructure)
- **Geological Feasibility Study:** Commission seismic surveys and salt dome analysis to determine suitability for cavern storage (if storage is contemplated)
- **Regulatory Pathway Assessment:** Determine whether FERC (interstate) or Louisiana PSC (intrastate) approval would be required; assess timeline and requirements

Deliverables

- Confirmed pipeline distances and system maps
- Preliminary geological assessment (salt dome feasibility)
- Regulatory pathway determination (FERC vs. PSC)
- Preliminary cost estimates for pipeline lateral
- Initial operator interest assessment

Timeline & Cost

6-9 months | $500K – $1.5M

Phase 2: Interconnection Feasibility & Commercial Framework

Objective

Define specific use case for pipeline access, develop commercial framework with pipeline operators, and secure experienced midstream partner.

Actions

- **Engineering Assessment:** Detailed route survey, pipeline sizing, interconnection design, cost estimation
- **Regulatory Requirements Evaluation:** FERC or Louisiana PSC application preparation; environmental and permitting analysis
- **Commercial Negotiations:** Preliminary discussions with pipeline operators regarding interconnection agreements, capacity rights, and commercial terms
- **Midstream Partnership Development:** Identify and engage experienced midstream partners with pipeline development and storage expertise
- **Market Demand Validation:** Assess whether storage, industrial supply, or other use case is economically justified by LNG load-balancing or regional demand

Deliverables

- Final engineering and cost estimates
- Draft interconnection agreements
- Regulatory application readiness
- Midstream partnership term sheet
- Market demand confirmation

Timeline & Cost

12-18 months | $3M – $8M

Decision Gate

Proceed to full development **only if Phase 2 confirms all four prerequisites**:

1. **Geological feasibility:** Salt dome suitable for cavern storage (if storage is the use case)
2. **Market demand:** Storage or industrial use justified by LNG load balancing or regional demand
3. **Operator partnership:** Experienced midstream partner committed
4. **Regulatory pathway:** FERC or Louisiana approvals achievable within acceptable timeline

> **GO / NO-GO DECISION**
>
> If any of the four prerequisites cannot be confirmed in Phase 2, **do not proceed to full development**. The capital intensity and regulatory complexity require all four conditions to be satisfied before committing to construction.

7. Conclusion

The Port Nickel site (387 acres, River Mile 80, Plaquemines Parish) occupies a strategic position within Southeast Louisiana's rapidly expanding energy infrastructure corridor. The site benefits from:

- **East Bank location** along established pipeline routes without river crossing requirements
- **Proximity to Gulf South Pipeline** (10-25 miles estimated), Louisiana's largest intrastate gas system
- **Positioning between two major demand centers:** New Orleans/Baton Rouge industrial corridor (north) and Plaquemines LNG terminals (south)
- **Same parish as largest U.S. LNG growth:** Plaquemines Parish hosts 70+ MTPA of LNG capacity (operational and under construction)
- **Existing midstream infrastructure presence:** Harvest CAM crude system (<5 miles) demonstrates infrastructure precedent

Southeast Louisiana is experiencing historic levels of pipeline infrastructure investment — **$5-10 billion in new natural gas transmission capacity added 2023-2027** — driven entirely by LNG export growth. Louisiana's LNG capacity is expanding from 15.3 Bcf/d (operational 2025) to 28+ Bcf/d by 2028, with Plaquemines Parish serving as the largest single concentration of growth.

This infrastructure boom creates potential opportunities for strategically located facilities that can serve as pipeline interconnection points, storage nodes, or industrial supply hubs. However, the capital intensity ($50M-$178M for pipeline connectivity alone, $220M-$658M with storage), regulatory complexity (FERC or Louisiana PSC approval required), and geological uncertainty (salt dome feasibility unconfirmed) necessitate a phased, disciplined approach.

Recommended Path Forward

1. **Phase 1 (6-9 months, $500K-$1.5M):** Obtain definitive pipeline location data from Louisiana DNR and operators; conduct preliminary geological assessment; determine regulatory pathway; establish initial operator interest
2. **Phase 2 (12-18 months, $3M-$8M):** Complete engineering and cost estimates; develop commercial framework with pipeline operators; secure experienced midstream partner; validate market demand
3. **Decision Gate:** Proceed to full development only if all four prerequisites are confirmed (geological feasibility, market demand, operator partnership, regulatory pathway)

The Port Nickel site's strategic location and Southeast Louisiana's infrastructure growth trajectory create potential value, but realization of that value depends on successful completion of Phase 1 and Phase 2

assessments. A disciplined, phased approach allows the project to be de-risked incrementally before committing significant capital.

PRIMARY OPPORTUNITY

Natural gas interconnection via Gulf South Pipeline system or other regional transmission infrastructure remains the most straightforward pathway. Proximity to Harvest CAM crude terminal and positioning along the East Bank pipeline corridor provide additional optionality for midstream development.

Next step: Initiate Phase 1 infrastructure mapping and preliminary assessment to convert proximity estimates into confirmed distances and establish operator engagement.

Annex A: Sources and Methodology

Primary Data Sources

- **FERC eLibrary and Major Pipeline Projects Database:** Pipeline certificate filings, environmental assessments, system maps (CP17-66, CP17-67, CP26-27)
- **U.S. EIA Natural Gas Infrastructure Data:** Pipeline capacity, throughput, regional flow patterns
- **Louisiana DNR Pipeline Program:** State pipeline registry and GIS data
- **Pipeline Operator Websites and Investor Presentations:** Boardwalk (Gulf South), Williams (Transco, LEG), Energy Transfer (Gulf Run), Momentum (NG3), DT Midstream (LEAP), TC Energy (Columbia Gulf), Venture Global (Plaquemines LNG, Gator Express, CP2)
- **RBN Energy Midstream Infrastructure Database:** Comprehensive pipeline inventory (180+ systems tracked)

Methodology

This analysis employed a multi-source verification approach to ensure accuracy:

1. **Pipeline Identification:** Cross-referenced FERC dockets, EIA data, Louisiana DNR records, and operator disclosures to identify all major pipeline systems in Southeast Louisiana
2. **Proximity Estimation:** Used publicly available system maps, FERC environmental filings, and regional infrastructure overlays to estimate distances from Port Nickel site. *Note: Estimates require confirmation via Louisiana DNR GIS data and direct operator engagement in Phase 1.*
3. **Capacity and Flow Data:** Sourced from EIA pipeline capacity reports, FERC certificate applications, and operator investor presentations
4. **LNG Terminal Data:** Compiled from FERC approvals, DOE export authorizations, and terminal operator announcements
5. **Market Analysis:** Based on EIA natural gas market reports, LNG export data, and Haynesville production forecasts

Limitations

- **Proximity estimates are preliminary** and require confirmation via Louisiana DNR pipeline GIS data and operator-provided system maps
- **Salt dome geology unconfirmed:** No seismic surveys or geological assessments conducted; storage feasibility requires Phase 1 investigation

- **Regulatory timeline estimates:** Based on historical FERC and Louisiana PSC approval timelines; actual timelines may vary
- **Cost estimates:** Order-of-magnitude ranges based on industry benchmarks; final costs depend on route selection, engineering, and site-specific factors

Annex B: Glossary of Terms

Term	Definition
Bcf/d	Billion cubic feet per day; standard unit for natural gas pipeline capacity and LNG terminal feedgas requirements
BPD	Barrels per day; standard unit for crude oil and refined products pipeline capacity
FERC	Federal Energy Regulatory Commission; regulates interstate natural gas pipelines, LNG terminals, and certain storage facilities
Interstate Pipeline	Natural gas pipeline crossing state boundaries; subject to FERC jurisdiction and open-access requirements
Intrastate Pipeline	Natural gas pipeline operating entirely within one state; subject to state regulation (Louisiana PSC); more flexible commercial terms
LNG	Liquefied Natural Gas; natural gas cooled to -260°F for export via tanker ships
MTPA	Million tonnes per annum; standard unit for LNG terminal capacity (1 MTPA ≈ 0.13-0.14 Bcf/d)
Haynesville Shale	Major natural gas producing formation in North Louisiana and East Texas; 15+ Bcf/d production; primary supply source for Louisiana LNG terminals
Gulf South Pipeline	Louisiana's largest intrastate natural gas transmission system; operated by Boardwalk Pipeline Partners; ~6 Bcf/d capacity
Transco	Transcontinental Gas Pipe Line; largest U.S. natural gas pipeline (~15 Bcf/d); operated by Williams Companies
Salt Cavern Storage	Underground natural gas or NGL storage in solution-mined salt dome caverns; provides rapid injection/withdrawal capability

Term	Definition
Lateral	Pipeline branch connecting a mainline transmission system to a specific facility, terminal, or interconnection point
Plaquemines LNG	Venture Global LNG export terminal in Plaquemines Parish, Louisiana; Phase 1 & 2 operational (3.25 Bcf/d / 20 MTPA); brownfield expansion proposed (30+ MTPA additional)
CP2	Venture Global's proposed LNG export terminal in Plaquemines Parish (adjacent to Plaquemines LNG); 3.25 Bcf/d / 20 MTPA; under construction
Louisiana PSC	Louisiana Public Service Commission; regulates intrastate pipelines, utilities, and certain energy infrastructure within Louisiana

Annex C: Pipeline Data Tables

Major Natural Gas Pipelines in Southeast Louisiana

Pipeline System	Operator	Type	Capacity	Est. Proximity	Key Route
Transcontinental (Transco)	Williams	Interstate	~15 Bcf/d	15-30 mi	Texas to Gulf Coast to Northeast
Texas Eastern (TETCO)	Enbridge	Interstate	~9 Bcf/d	25-40 mi	Texas/Louisiana to Northeast
Tennessee Gas Pipeline (TGP)	Kinder Morgan	Interstate	~9 Bcf/d	20-35 mi	Gulf Coast to Northeast
Gulf South Pipeline	Boardwalk	Intrastate	~6 Bcf/d	10-25 mi	Louisiana intrastate network
Columbia Gulf	TC Energy	Interstate	~3.1 Bcf/d	30-50 mi	Gulf Coast to Midwest
Louisiana Energy Gateway (LEG)	Williams/Enbridge	Intrastate	1.44 Bcf/d	30+ mi	Haynesville to Cameron LNG
Gulf Run	Energy Transfer	Intrastate	1.65 Bcf/d	30+ mi	Haynesville to Golden Pass LNG
LEAP	DT Midstream	Intrastate	1.15 Bcf/d	30+ mi	Haynesville to Plaquemines area
NG3	Momentum Midstream	Intrastate	1.65 Bcf/d	30+ mi	Haynesville to CP2

Major Crude Oil and Products Systems

System	Operator	Type	Capacity	Route/Terminals
LOOP	LOOP LLC	Crude Import	1.2M BPD	Offshore to Clovelly to St. James
Genesis Offshore	Genesis Energy	Crude Gathering	400K+ BPD	GOM platforms to shore
Harvest CAM	Harvest Midstream	Crude Gathering	N/A	Belle Chasse terminal area
Colonial Pipeline	Colonial Pipeline Co.	Products	2.5M BPD	Gulf Coast to Southeast U.S.
Plantation	Kinder Morgan	Products	720K BPD	Louisiana to Washington DC
Collins	Marathon/Valero	Products	190K BPD	South Louisiana distribution

Louisiana LNG Export Terminals

Terminal	Operator	Parish	Capacity (Bcf/d)	Capacity (MTPA)	Status
Sabine Pass	Cheniere	Cameron	4.2	27.0	Operational
Cameron LNG	Sempra/TotalEnergies/Mitsui	Cameron	2.45	15.8	Operational (Train 4 added 2024)
Calcasieu Pass (CP1)	Venture Global	Cameron	2.6	16.8	Operational
Plaquemines LNG	Venture Global	Plaquemines	3.25	20.0	Operational (2024-2025)
Golden Pass LNG	ExxonMobil/Qatar Energy	Jefferson (TX)	2.6	16.8	Operational (2024-2025)
CP2	Venture Global	Plaquemines	3.25	20.0	Under construction
Plaquemines Brownfield	Venture Global	Plaquemines	4.6+	30.0+	FERC filing (Nov 2025)

Total Operational Louisiana LNG Capacity (2025): 15.3 Bcf/d

Total Under Construction/Approved: 12.7+ Bcf/d

Projected Total by 2030: 28+ Bcf/d

Annex D: References

Government and Regulatory Sources

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Document Control

Field	Value
Document Title	Southeast Louisiana Pipeline Infrastructure — Port Nickel Comprehensive Analysis
Project	MASABB Site Assessment
Site Location	387 Acres, Mississippi River Mile 80 (East Bank), Braithwaite, Plaquemines Parish, Louisiana
Prepared For	Port Nickel Development Team
Prepared By	William S. Davis III
Organization	SESCO Cement Internal Document
Date	February 5, 2026
Revision	1.0 - Final Comprehensive Report
Document Type	Infrastructure Analysis / Site Assessment
Classification	Internal
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1.0	February 5, 2026	Final comprehensive merged report	W.S. Davis III

Distribution

- Port Nickel Development Team
- SESCO Cement Management
- Project File

END OF REPORT

Southeast Louisiana Pipeline Infrastructure — Port Nickel Comprehensive Analysis

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